Ex.No. 3.1 – Design and Draft the Given 3D Component

AIM

To create the 3D model and dimensioned drawing of the given forked component using Autodesk Fusion 360.

PROCEDURE

1. Open Autodesk Fusion 360, create a new design, and set the document units to millimetres.

2. On the XY plane, sketch the main rectangular body using the drawing dimensions: 40 mm overall width and 40 mm height in the front view. Extrude the profile to form the main block, keeping the overall body length as 52 mm.

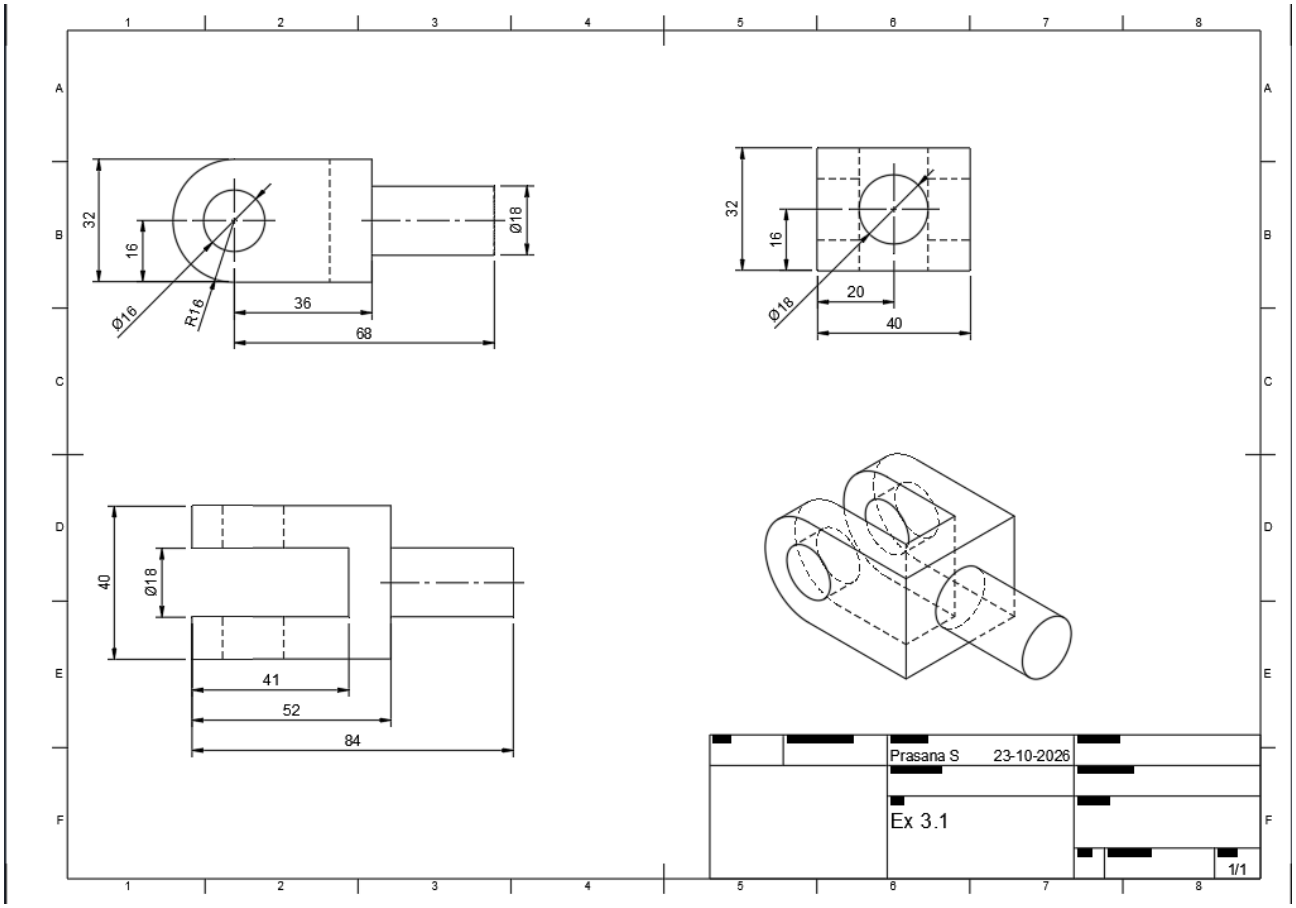
3. Create the forked end by sketching the central opening. Set the opening length to 41 mm and the slot width to 18 mm, then cut it through the required portion of the body to form the two parallel lugs.

4. At the rounded end of each lug, create concentric circles of Ø18 mm and Ø16 mm as indicated in the drawing. Cut the Ø16 mm holes through both lugs, keeping them aligned.

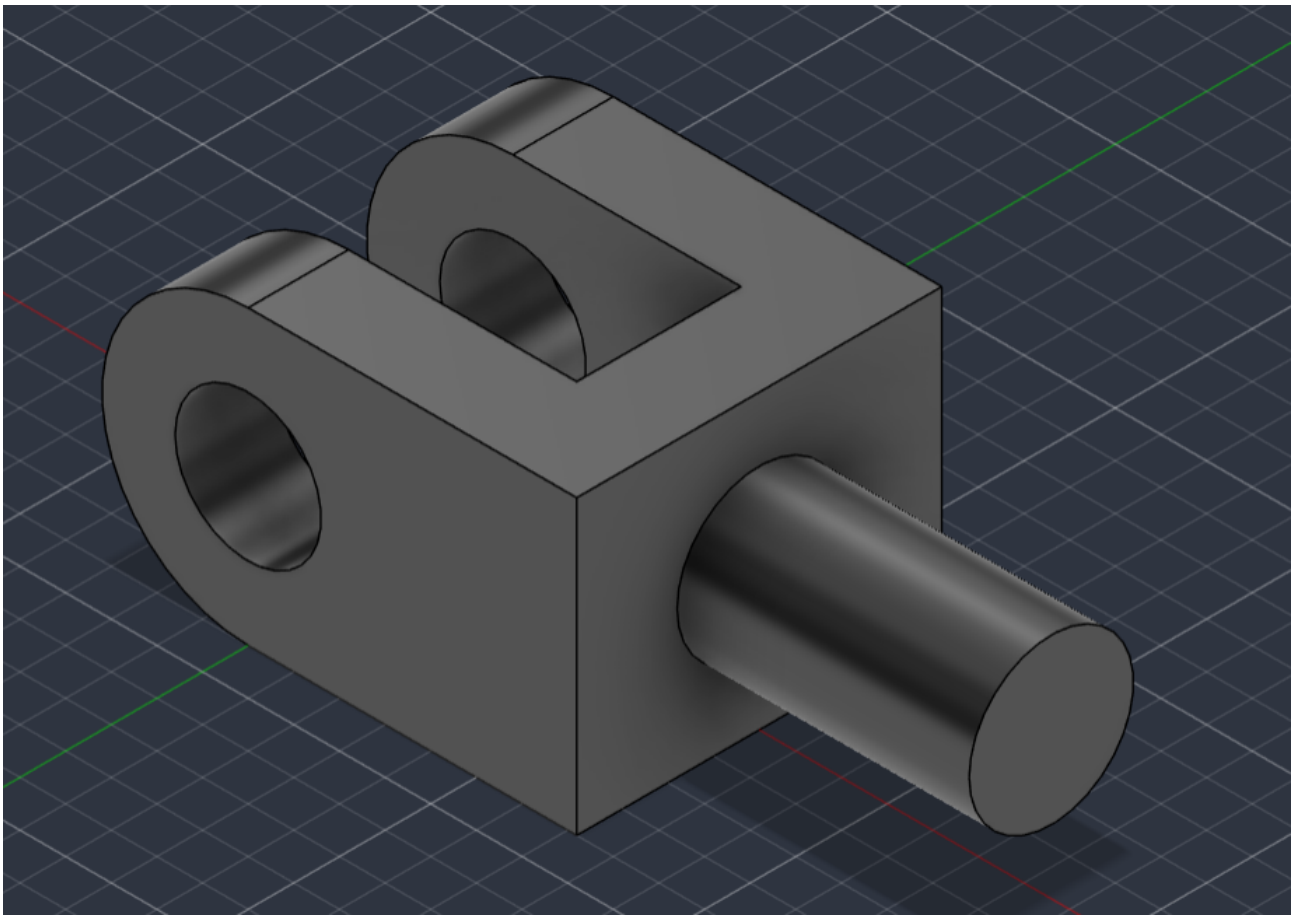
5. On the opposite face, sketch a coaxial cylinder of Ø18 mm and create the projecting shaft. Position it on the centre line and set the overall length from the drawing to 84 mm.

6. Check the principal dimensions—40 mm body width/height, 52 mm body length, 41 mm fork opening length, Ø18 mm lug outside profile, Ø16 mm lug holes, and Ø18 mm shaft—and save the model.

SKETCHES



OUTPUT



RESULT

Thus, the given component was modelled and drafted using Autodesk Fusion 360.